

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – Industry Problem Solving Task

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO4 – Precision	Assessment:	Assessment Tool 2 – Industry Problem Solving Task
Weightage:	35% of CIA		
CO5 Statement:	Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4)		
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Section / Batch:	2024	Date:	

Code of Conduct

I Mohamed Jameel A/192425057 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Instructions

- Read each industry scenario carefully before answering.
- Analyze the given problem from a semantic analysis perspective.
- Provide structured and logical answers with proper justification.
- Support your analysis using examples from the given scenario wherever applicable.
- Use suitable diagrams, semantic representations, logical predicates, workflows, architectures, or tables whenever necessary.
- Clearly state assumptions if additional information is required.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
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Representing Meaning & Meaning Structure of Language	Analyze semantic interpretation of user queries, identify ambiguity in meaning representation, evaluate semantic processing limitations, and improve understanding in banking chatbot applications.	Q1
Syntax-Driven Semantic Analysis	Analyze multiple interpretations of spoken commands, evaluate parsing-related ambiguity, and assess semantic extraction in real-time voice assistant systems.	Q2
First-Order Predicate Calculus, Representing Linguistically Relevant Concepts, Syntax-Driven Semantic Analysis	Design a semantic processing architecture for healthcare reports, model relationships among entities and actions, represent linguistic concepts, and improve semantic extraction accuracy.	Q3

Annexure A – Industry Problem Solving Task

1. AI-Powered Insurance Claim Processing System

An insurance company is developing an NLP-based system to automatically process customer insurance claims. The system uses **Context-Free Grammar (CFG)** to parse customer statements and extract information such as the claimant, damaged object, cause of damage, and claim amount.

However, the system faces the following problems:

- Ambiguous sentence structures
- Difficulty distinguishing noun and prepositional phrase attachment
- Subject–verb agreement errors
- Inefficient parsing of lengthy customer descriptions
- Difficulty handling conversational and incomplete input

Example customer statement:

"I reported the damage to the car after the accident."

The sentence can produce different interpretations depending on whether "**to the car**" is associated with *reported* or *damage*.

Tasks:

- a) Analyze the structural ambiguity in the given sentence using CFG and illustrate the possible interpretations.
- b) Evaluate the limitations of a basic CFG in handling ambiguity, agreement, and long customer-generated insurance statements.
- c) Design an improved parsing approach using **PCFG, Feature Structures, and Earley Parsing** to resolve ambiguity and efficiently process the claim.
- d) Justify how the proposed architecture can improve the accuracy, scalability, and reliability of automated insurance claim processing.

2. Intelligent Railway Ticket Booking Assistant

A railway company is developing a conversational AI assistant for ticket booking. The system currently uses **top-down parsing** to interpret passenger requests.

The system encounters:

- Excessive backtracking
- Ambiguous attachment of phrases
- Incomplete passenger requests
- Long-distance dependencies
- Slow response for complex queries

Example passenger request:

"Book two tickets from Chennai to Delhi for my parents with AC sleeper."

The system sometimes incorrectly associates "**with AC sleeper**" with the passengers instead of the ticket/class information.

Tasks:

- Analyze the possible syntactic interpretations of the given passenger request using CFG.
- Evaluate why top-down parsing may result in excessive backtracking and inefficient processing for conversational ticket-booking queries.
- Analyze how **Earley Parsing** can handle ambiguity, incomplete input, and different possible parse structures more effectively.
- Compare **Top-Down Parsing, CFG-based parsing, and Earley Parsing** in terms of ambiguity handling, computational efficiency, partial-input processing, and suitability for real-time railway booking systems.

3. AI-Based Legal Contract Analysis System

A legal technology company is developing an NLP system to analyze contracts and automatically identify:

- Parties involved
- Obligations
- Conditions
- Deadlines
- Actions
- Exceptions

The existing system uses a basic CFG parser. However, it produces incorrect interpretations when processing long legal sentences containing relative clauses, passive constructions, and multiple prepositional phrases.

Example contract sentence:

"The supplier who delivered the equipment to the company must replace the defective components within thirty days."

The system sometimes incorrectly identifies the **company** as the entity responsible for delivering the equipment.

Tasks:

- a) Analyze the syntactic structure of the given sentence using CFG and identify possible sources of ambiguity.
- b) Evaluate why basic CFG is insufficient for accurately interpreting complex legal sentences containing relative clauses and nested structures.
- c) Design an NLP architecture integrating **CFG, PCFG, Feature Structures, and Earley Parsing** to improve syntactic and semantic interpretation.
- d) Develop a step-by-step processing workflow showing how the system can extract:
 - Supplier
 - Action
 - Equipment
 - Company
 - Obligation
 - Deadline

and justify how the proposed approach improves automated contract analysis.

4. AI-Based E-Commerce Product Search System

An e-commerce company is developing a conversational product-search system. Customers enter natural-language queries rather than selecting predefined filters.

The current system uses CFG but faces difficulties with:

- Ambiguous phrase attachment
- Coordination
- Missing words
- Informal customer queries
- Long product descriptions
- Real-time search requirements

Example user query:

"Find laptops with a touchscreen for students under ₹60,000."

The system may incorrectly associate **"for students"** with the touchscreen rather than the laptop.

Tasks:

- a) Analyze the syntactic ambiguity in the given query and construct possible parse interpretations using CFG.
 - b) Evaluate the limitations of CFG in handling informal, elliptical, and ambiguous e-commerce search queries.
 - c) Propose an improved parsing architecture using **PCFG, Feature Structures, and Earley Parsing** to identify product attributes, user requirements, and price constraints.
 - d) Justify how the proposed solution can improve search precision, response time, and customer experience in a large-scale e-commerce platform.
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5. Intelligent Airline Voice Assistant

An airline is developing a real-time voice assistant that allows passengers to modify flight bookings using spoken commands.

The system currently uses a conventional top-down parser. It experiences:

- Backtracking
- Ambiguous attachment
- Speech recognition errors
- Incomplete commands
- Delayed responses
- Difficulty interpreting corrections made by users

Example spoken command:

"Change my flight to Mumbai tomorrow with two passengers."

Depending on the parse, **"with two passengers"** may be incorrectly interpreted as an attribute of the destination rather than the number of passengers.

The passenger may also say:

"Change my flight to Mumbai tomorrow... actually, make that Bangalore."

Tasks:

- a) Analyze the possible syntactic structures and ambiguities in the given voice command.

- b)** Evaluate the limitations of top-down parsing when processing incomplete, corrected, and ambiguous spoken commands in real time.
- c)** Design a parsing architecture using **Earley Parsing, PCFG, and Feature Structures** to support partial input, ambiguity resolution, and agreement constraints.
- d)** Develop a processing workflow showing how speech input can be converted into a structured booking request and justify the proposed method for real-time airline applications.

Rubrics for Calculation

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding ; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution

Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

Q. No. / Criteria Level	Understanding of Industry Problem	Application of Engineering Concepts	Solution Design	Use of Tools	Analysis	Professionalism	Sub. Total	Total %
1								
2								
3								
4								
5								

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____

Date: _____	Date: _____	Date: _____
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Q1 — AI-Powered Insurance Claim Processing System

a) Structural Ambiguity using CFG

Given sentence:

“I reported the damage to the car after the accident.”

The phrase **“to the car”** can attach to different parts of the sentence, producing different interpretations.

Interpretation 1 — to the car modifies damage

I reported [the damage [to the car]] [after the accident].

Meaning:

Damage → to the car

Action → reported

Time → after the accident

The claimant is reporting **damage that occurred to the car**.

Interpretation 2 — to the car modifies reported

I reported [the damage] [to the car] [after the accident].

This gives the less likely interpretation that the reporting/action is somehow directed **to the car**.

Simplified CFG

$S \rightarrow NP VP$

$VP \rightarrow V NP PP$

$VP \rightarrow V NP PP PP$

$NP \rightarrow Det N$

$PP \rightarrow P NP$

The ambiguity arises because the PP **“to the car”** can be attached to different constituents.

b) Limitations of Basic CFG

A basic CFG has several limitations for this insurance system:

- **Ambiguity:** Multiple valid parse trees may be generated for the same sentence.
- **PP attachment:** It cannot reliably determine which noun/verb a prepositional phrase belongs to.
- **Agreement:** Basic CFG does not naturally enforce features such as singular/plural subject–verb agreement.
- **Long descriptions:** Customer-generated insurance statements can produce many possible

parses, increasing computational cost.

- **Conversational/incomplete input:** CFG expects relatively well-formed grammatical structures and may struggle with fragments or informal claims.

c) Improved Parsing Approach

Use a combination of **PCFG + Feature Structures + Earley Parsing**.

Customer Statement



Tokenization



Feature Structure Analysis



Earley Parser



Multiple Candidate Parse Trees



PCFG Probability Ranking



Best Parse



Semantic Role / Claim Information Extraction



Insurance Claim

Role of each technique

Technique	Purpose
CFG	Defines the basic grammatical structure
Feature Structures	Handles agreement and grammatical constraints
Earley Parser	Efficiently handles ambiguity and incomplete input
PCFG	Assigns probabilities to possible parses and selects the most likely interpretation

For this sentence, PCFG can rank the interpretation where **“to the car” describes the damage** higher than the unlikely alternative.

d) Why the Architecture Improves the System

Accuracy

PCFG helps select the **most probable interpretation**, while feature structures reduce grammatically invalid parses.

Scalability

Earley parsing can efficiently handle complex and ambiguous customer statements compared with repeatedly exploring possibilities through basic top-down parsing.

Reliability

Combining syntactic parsing with semantic extraction ensures the system correctly identifies important claim information such as:

Claimant → I

Damage → damage to the car

Cause → accident

Time → after the accident

This reduces the chance of sending an incorrectly interpreted claim for automated processing.

Conclusion

The proposed **CFG + Feature Structures + Earley + PCFG** architecture is more suitable for an industry insurance system because it addresses the exact problems identified in the scenario—**ambiguity, agreement, lengthy descriptions, and conversational input**—while producing a ranked and semantically useful interpretation.

Q2 — Intelligent Railway Ticket Booking Assistant

a) Possible CFG Interpretations

Given:

“Book two tickets from Chennai to Delhi for my parents with AC sleeper.”

The main ambiguity is the attachment of **“with AC sleeper.”**

Interpretation 1 — Ticket/Class

Book [two tickets from Chennai to Delhi
for my parents] [with AC sleeper].

Meaning:

Passengers → my parents

Number → 2

From → Chennai

To → Delhi

Class → AC sleeper

Interpretation 2 — Passenger-related

Book [two tickets from Chennai to Delhi
for my parents with AC sleeper].

Here, the parser may incorrectly associate **“with AC sleeper”** with the passengers rather than the ticket/class information.

The correct interpretation is **Interpretation 1**.

b) Problems with Top-Down Parsing

The current top-down parser can struggle because it:

- Performs **excessive backtracking** when multiple grammar rules are possible.
- Has difficulty resolving **ambiguous phrase attachment**.
- Struggles with **incomplete passenger requests**.
- Has difficulty with **long-distance dependencies**.
- Can become slow for complex conversational queries.

For example, it may explore several ways of attaching:

"with AC sleeper"

before finding the correct interpretation.

c) How Earley Parsing Helps

Earley Parsing is better suited because it can maintain **multiple possible parses** while processing the sentence rather than repeatedly restarting through excessive backtracking.

Passenger Request



Tokenization



Earley Parser



Possible Parse Structures



PCFG / Semantic Ranking



Correct Interpretation



Booking Information

It is particularly useful for:

- **Ambiguous input**
- **Incomplete input**
- **Different possible parse structures**

Thus, the system can preserve alternative interpretations until sufficient context is available.

d) Comparison

Feature	Top-Down Parsing	CFG-Based Parsing	Earley Parsing
Ambiguity	Can cause backtracking	Can generate multiple parses	Handles multiple parses efficiently
Efficiency	Can become slow with backtracking	Depends on parsing method	More suitable for complex input
Partial input	Limited	Limited	Better support
Conversational queries	Less suitable	Moderate	More suitable
Real-time booking	Can cause delays	Basic solution	Better choice

Conclusion

For a railway booking assistant, **Earley Parsing is the most suitable of the three** because the system must handle ambiguous, incomplete, and conversational passenger requests while maintaining acceptable response time.

Q3 — AI-Based Legal Contract Analysis System

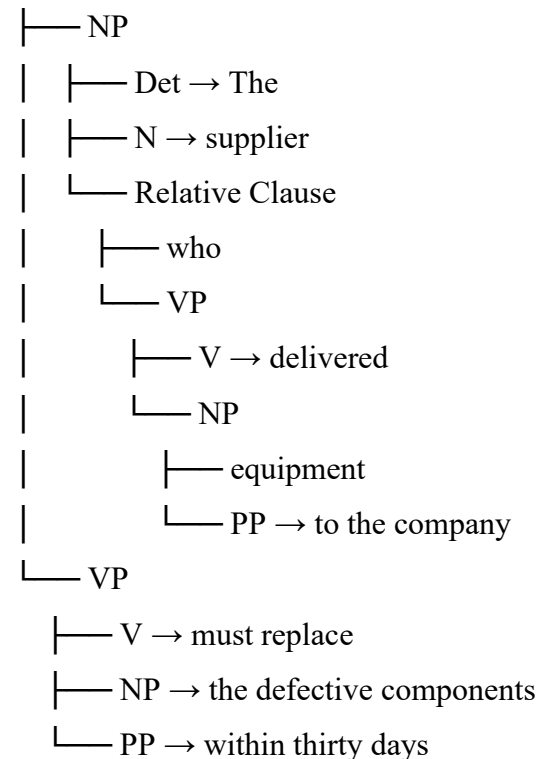
a) CFG Structure and Ambiguity

Sentence:

“The supplier who delivered the equipment to the company must replace the defective components within thirty days.”

A simplified CFG structure is:

S



Important relationship

supplier → delivered → equipment

company → recipient/destination

supplier → must replace → defective components

deadline → within thirty days

The main ambiguity comes from the **relative clause** and its embedded prepositional phrase:

who delivered [the equipment] [to the company]

A basic parser may incorrectly associate **“the company”** with the entity performing the delivery, which is why the system incorrectly identifies the company as responsible.

b) Why Basic CFG Is Insufficient

A basic CFG is limited because:

- It can generate multiple possible structures without reliably selecting the intended one.
- It does not naturally capture **semantic relationships** between entities.
- Nested **relative clauses** increase parsing complexity.
- Multiple **prepositional phrases** can create attachment ambiguity.
- Long legal sentences can contain deeply nested structures.

Therefore, CFG alone cannot reliably determine that **supplier** is the entity who delivered the equipment and who has the obligation to replace it.

c) Improved NLP Architecture

Legal Contract



Sentence / Clause Segmentation



CFG Parsing



Earley Parsing



Feature Structure Analysis



PCFG Ranking



Semantic Role Analysis



Entity & Obligation Extraction



Structured Contract Information

Role of each component

Component	Purpose
CFG	Defines grammatical structures
Earley Parser	Handles complex/nested and ambiguous structures
Feature Structures	Maintains grammatical constraints and relationships
PCFG	Ranks competing interpretations using probabilities
Semantic Analysis	Maps entities to actions, obligations and roles

This combination directly addresses the limitations identified in the scenario.

d) Processing Workflow

Step 1 — Identify entities

Supplier

Equipment

Company

Defective components

Step 2 — Identify actions

Supplier → delivered → Equipment

Supplier → must replace → Defective components

Step 3 — Identify relationships

Company → destination/recipient of delivery

Step 4 — Identify obligation

Supplier

↓ must replace

Defective components

Step 5 — Identify deadline

Deadline = within 30 days

Final Structured Representation

Supplier = supplier

Action 1 = delivered

Equipment = equipment

Company = recipient/destination

Action 2 = replace

Obligation = supplier must replace defective components

Deadline = 30 days

Conclusion

The proposed **CFG + PCFG + Feature Structures + Earley Parsing** architecture improves contract analysis by resolving complex syntactic structures before semantic extraction. It correctly identifies the **supplier as the entity responsible for both delivery and replacement**, while extracting the equipment, company, obligation, and 30-day deadline.

Q4 — AI-Based E-Commerce Product Search System

a) Syntactic Ambiguity

Query:

“Find laptops with a touchscreen for students under ₹60,000.”

The ambiguity is mainly the attachment of **“for students.”**

Interpretation 1 — Correct

Find [laptops [with a touchscreen] [for students] [under ₹60,000]]

Meaning:

Product → laptops

Attribute → touchscreen

Target users → students

Price limit → ₹60,000

Interpretation 2 — Incorrect

Find [laptops [with a touchscreen for students] [under ₹60,000]]

Here, **“for students”** may incorrectly be treated as describing the touchscreen rather than the laptops.

The system should select **Interpretation 1**.

b) Limitations of CFG

A basic CFG may struggle with:

- **Ambiguous phrase attachment**

- **Coordination**
- **Missing words / elliptical queries**
- **Informal customer language**
- **Long product descriptions**
- **Real-time search requirements**

CFG can represent grammatical structures, but it does not inherently know the user's intended **product, attributes, requirements, and constraints.**

c) Improved Parsing Architecture

Customer Query



Tokenization



CFG Parsing



Earley Parser



Feature Structure Analysis



PCFG Ranking



Semantic Attribute Extraction



Product Search



Ranked Results

Role of each technique

Technique	Purpose
CFG	Represents basic syntactic structures
Earley Parsing	Handles ambiguous and partial input

Technique	Purpose
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Feature Structures	Represents constraints and relationships
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PCFG	Ranks competing interpretations
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The resulting semantic representation can be:

Product = Laptop

Feature = Touchscreen

User = Students

Price $\leq$ = ₹60,000

This directly addresses the assessment's requirement to identify **product attributes, user requirements, and price constraints**.

d) Benefits to E-Commerce Search

Search Precision

Correctly separating:

Laptop → Product

Touchscreen → Attribute

Students → User Requirement

₹60,000 → Price Constraint

prevents the search engine from misinterpreting the query.

Response Time

Earley parsing avoids unnecessary repeated parsing of ambiguous structures, while PCFG helps select the most likely interpretation.

Customer Experience

The system can understand natural conversational queries instead of forcing customers to use predefined filters.

Conclusion

A **CFG + PCFG + Feature Structures + Earley Parsing** architecture can provide more accurate semantic interpretation of natural-language product queries, improving **search precision, response time, and overall customer experience**.

Q5 — Intelligent Airline Voice Assistant

a) Syntactic Structures and Ambiguity

Command:

“Change my flight to Mumbai tomorrow with two passengers.”

Correct interpretation

Change

|— Flight → my flight
|— Destination → Mumbai
|— Date → tomorrow
|— Passengers → two

So:

Destination = Mumbai

Date = tomorrow

Passengers = 2

Ambiguity

The phrase **“with two passengers”** may incorrectly be attached to **Mumbai**, instead of the booking/passenger information.

There is also a correction:

“Change my flight to Mumbai tomorrow... actually, make that Bangalore.”

The final destination should therefore be:

Destination = Bangalore

because the user explicitly corrects Mumbai to Bangalore.

b) Limitations of Top-Down Parsing

A conventional top-down parser can struggle with:

- **Backtracking** when multiple interpretations exist.
- **Ambiguous phrase attachment.**
- **Incomplete spoken commands.**
- **Speech recognition errors.**
- **User corrections** during conversation.
- **Delayed responses** for complex inputs.

For example:

"Change my flight to Mumbai tomorrow...

actually, make that Bangalore."

The system must understand that **Bangalore replaces Mumbai**, rather than treating it as an additional destination.

c) Improved Parsing Architecture

Speech Input



Speech Recognition



Tokenization



Earley Parser



Multiple Parse Structures



Feature Structures



PCFG Ranking



Correction / Context Handling



Semantic Representation



Booking System

Role of Components

Component	Purpose
Earley Parser	Handles partial and ambiguous input

Component	Purpose
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PCFG	Ranks possible interpretations
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Feature Structures	Maintains constraints such as date, destination and passenger count
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Correction handling	Updates previously extracted information
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The architecture directly addresses the assessment requirements for **partial input, ambiguity resolution, and agreement constraints**.

d) Processing Workflow

1. Speech → Text

"Change my flight to Mumbai tomorrow with two passengers."

2. Extract information

Action → Change Flight

Destination → Mumbai

Date → Tomorrow

Passengers → 2

3. Process correction

"Actually, make that Bangalore."

Update:

Destination: Mumbai → Bangalore

4. Final Structured Request

Action = Change Flight

Destination = Bangalore

Date = Tomorrow

Passengers = 2

5. Send to Booking System

Final Request



Validate details



Modify flight



Confirm with passenger

Conclusion

The proposed **Earley + PCFG + Feature Structures** architecture is more suitable for a real-time airline assistant because it can maintain alternative interpretations, process incomplete speech, enforce constraints, and incorporate user corrections before sending the final structured request to the booking system.